

# AYMANE AIT DADS

AI Research & Development Intern | LLM Fine-Tuning · Computer Vision · Retrieval-Augmented Generation  
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## PROFESSIONAL SUMMARY

Achieved top 10% on the NVIDIA Nemotron Reasoning Kaggle competition (0.80/1.0, \$106K+ prize pool) by fine-tuning a 30B Mamba-Transformer MoE model with LoRA on chain-of-thought reasoning tasks, directly demonstrating LLM research and rigorous benchmarking. Ranked 1st in EURECOM class on an AI-generated image detection benchmark (0.791 AUC) using CLIP ViT-L/14, with a concurrent submission to the NTIRE 2026 @ CVPR CodaBench international benchmark, validating prototype-to-benchmark research methodology. Proficient in Large Language Models, Retrieval-Augmented Generation, Computer Vision, PyTorch, Python, Docker, Git, and Linux environments, with hands-on experience conducting ablation studies, data preparation, and model evaluation across academic and competition settings. Available for a 6-month full-time internship starting 2025, with flexibility on dates, and currently exploring work authorization pathways for Singapore.

## CORE COMPETENCIES

Large Language Models (LLMs) | Retrieval-Augmented Generation | Computer Vision | Model Benchmarking & Evaluation | Proof-of-Concept Prototyping | Deep Learning | Data Preparation | Git & Linux Environments

## WORK EXPERIENCE

### Orange Maroc

Jul 2024 - Aug 2024

**Data Science Intern** - Casablanca, Morocco

- Designed and validated machine learning models (Random Forest + K-Means) on **100K+ fiber-optic network records**, delivering rigorous evaluation and benchmarking of network quality across thousands of nodes.
- Developed an end-to-end clustering pipeline that mapped customer performance profiles to **5 commercial service tiers**, producing proof-of-concept output directly adopted by the network optimization team.
- Reduced manual analysis time by **~60%** through automated feature engineering and model evaluation pipelines built in Python, Pandas, and Scikit-learn.
- Translated model outputs into stakeholder presentations with actionable maintenance recommendations, demonstrating **communication and analytical problem-solving** across technical and business audiences.

## PROJECTS

### NVIDIA Nemotron Reasoning Challenge

Kaggle Competition · In Progress, June 2026 · Top 10% Public Leaderboard

- Achieved **top 10% public leaderboard score (0.80/1.0, \$106K+ prize pool)** by fine-tuning a 30B Mamba-Transformer MoE model with LoRA ( $r=32$ , ~888M trainable parameters) on 7,828 chain-of-thought reasoning examples using SFTTrainer with completion-only loss.
- Resolved **5 Blackwell GPU (sm\_120) incompatibilities** in strict order - including OOM patching, Mamba CUDA kernel disabling, and Triton fallbacks - and conducted 20+ single-variable ablations confirming that packing and synthetic data consistently degraded score by up to 0.30.

PyTorch · Unsloth · PEFT · Hugging Face Transformers · vLLM · TRL · Triton

### AI-Generated Image Detection

EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in Class

- Ranked **1st in EURECOM class (private Kaggle leaderboard, 0.791 AUC)** by fine-tuning CLIP ViT-L/14 (428M params) with a custom MLP head, dual learning rates, and a 3-model ensemble with 10-view test-time augmentation across 250K images from 25+ generator types.
- Submitted to the **NTIRE 2026 @ CVPR CodaBench** international benchmark, validating prototype-to-production research methodology; key finding showed 1-epoch training (0.793 AUC) outperformed 4-epoch (0.767 AUC) due to out-of-distribution generalization.

PyTorch · OpenCLIP · ViT · Hugging Face Transformers · Kaggle T4 GPU

### Agentic LLM Tooling & RAG Prototyping

Independent · 2024-2025 · Applied AI Systems

- Built LLM orchestration prototypes using **Claude API, LangChain, and RAG pipelines** with Supabase and PostgreSQL backends, integrating MCP server tool-calling and REST APIs for end-to-end agentic workflows.
- Applied **prompt engineering, retrieval-augmented generation, and embedding-based search** to develop proof-of-concept systems demonstrating feasibility for real-world enterprise AI applications.

LangChain · Claude API · RAG · Supabase · PostgreSQL · REST APIs

## EDUCATION

### Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Image Security, Cloud Computing, Statistics, Web Applications

### CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

## SKILLS

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**LLM Fine-Tuning & NLP:** LoRA / QLoRA | SFTTrainer | Hugging Face Transformers | RAG | LangChain | Prompt Engineering | Chain-of-Thought

**Computer Vision & Deep Learning:** PyTorch | CLIP / ViT | CNN / DenseNet | TensorFlow | Scikit-learn | OpenCLIP | FP16 Mixed Precision

**MLOps & Infra:** Python | Git | Linux | Docker | Supabase / PostgreSQL | Power BI | CUDA Debugging